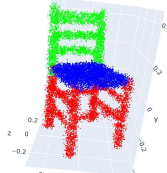


1. Multimodal Observation

RGB Image



3D Geometry



Language Instruction

'Pick up the wooden chair and move it to the left.'



2. Object-centric Representation

Shape



Semantic



Physical Cues



Size / Mass



Material Hint



Physical Context



3. Physical Understanding

Appearance



color, texture, roughness)

Material & Mass



(density, stiffness, mass)

Dynamics



(friction, deformation, damping)

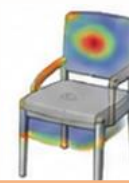
Physically Grounded Properties

4. Manipulation Affordance

Grasp



Push



Pull



Lift/Place



...

Actionable Affordance Map & Modes

5. Actionable Object Representation



Geometry



Semantics



Physical Properties



Affordance Modes

Robot Manipulation

